



CONDITION BASED ASSESSMENT

FULL SCANNING REPORT

SBUM DIAGNOSTIC PAHANG

TNB BAHAGIAN PEMBAHAGIAN PAHANG

FUNCTIONAL LOCATION (ERMS)	:	CKTN/PCE/J01028
FUNCTIONAL LOCATION (SITE)	:	CKTN/PCE/J01028
SUBSTATION NAME (ERMS)	:	CS LOT 7011 DARAT MAKBAR
SUBSTATION NAME (SITE)	:	CS LOT 7011 DARAT MAKBAR
COORDINATE	:	3.809330, 103.318326
SUBSTATION TYPE	:	CS
AREA	:	KUANTAN
STATE	:	PAHANG
PO	:	42225439
DATE	:	03 MAR 2026

SCANNED BY	
THERMOGRAPHY	: MUHAMMAD ADAM BIN SAFARUDDIN (2017MY34N004)

EQUIPMENT/TOOLS	
THERMOGRAPHY	: FLIR T640 (SN 55912229) (CALIBRATION DUE DATE: 26 DEC 2026)

PREPARED BY	
NAME OF CONTRACTOR	: EET ENGINEERING SERVICES SDN BHD
ADDRESS	: NO.38, JALAN SS19/1D,
	: 47500, SUBANG JAYA, SELANGOR.

SUBSTATION SIGNBOARD



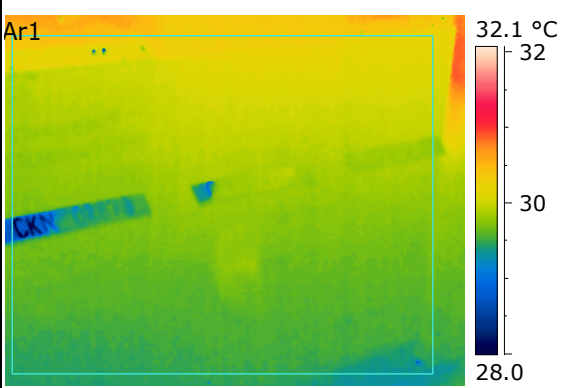
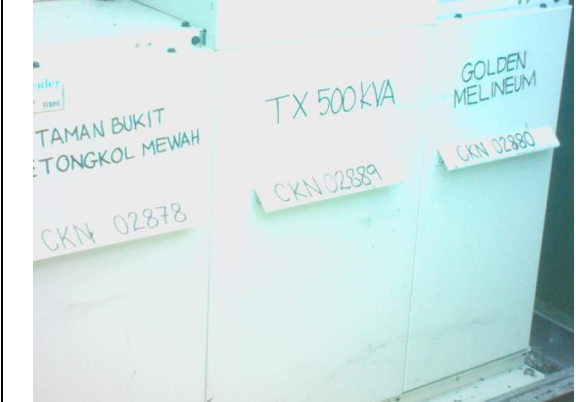
EXECUTIVE SUMMARY

NO.	EQUIPMENT	DEFECT AREA	IR (Abs T/ Δ T)	DEFECT
1.	MRMU, SCHNEIDER	OVERVIEW	-	-
		OVERVIEW TOP	-	-
	PANEL CKN02878 TAMAN BUKIT SETONGKOL MEWAH	CABLE COMPARTMENT	-	
	PANEL CKN02889 TX 500KVA	CABLE COMPARTMENT	-	
	PANEL CKN02880 GOLDEN MELINEUM	CABLE COMPARTMENT	-	
2.	TX1 - EWT 500KVA	HV BUSHING	-	
		HV CABLES	-	
3.	LVDB TX1	OVERVIEW	-	-
4.	BATTERY 1	OVERVIEW	-	-

VISUAL DEFECT SUMMARY

NO.	EQUIPMENT	DEFECT DESCRIPTION	ADDITIONAL REMARKS
1	SUBSTATION	-	PADLOCK HOLDER BROKEN - LVDB TX1 BAY
2	TRENCHING	NO COVER	SWG BAY
3	RUBBER MAT	NO RUBBER MAT	SWG BAY
4	EFI	NO BATTERY SUPPLY	CKN02880 - GOLDEN MELINEUM
5	FP/LVDB	NO FEEDER NO./FEEDER NAME	LVDB TX1 - NOT FOLLOW NEW STANDARD

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	MRMU	Manufacturer	SCHNEIDER	Time: 09:28 AM
	Model	RM6	Rating	12kV	Humidity: 77%
	Area	OVERVIEW		Panel No.	-
Panel Name		-			

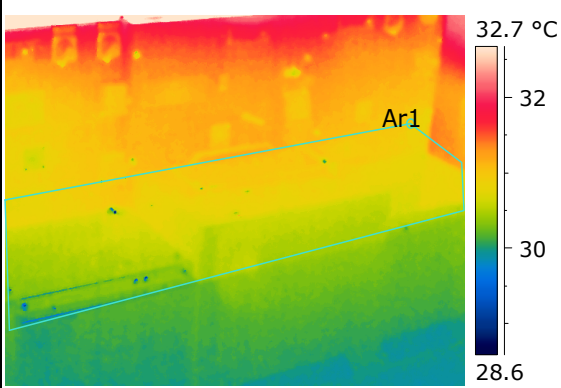
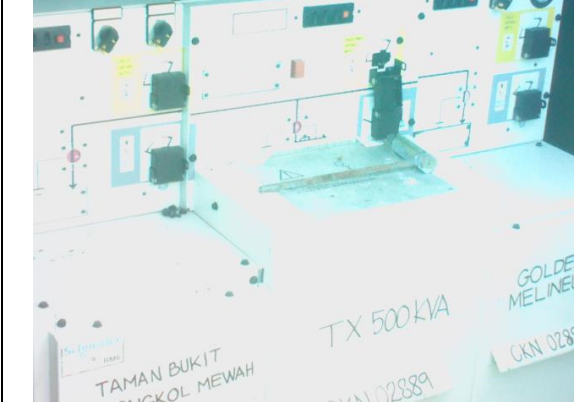
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	-	Serial No.	MA223850062
Spot Temp. 2 (Sp2)	-	Heater (A)	-
Average Temp. (Ar1)	29.8 °C	CB Position	-
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (ΔT)	-	Cable Type	-
Load (A)	-		
Defect	-		

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	MRMU	Manufacturer	SCHNEIDER	Time: 09:28 AM
	Model	RM6	Rating	12kV	Humidity: 77%
	Area	OVERVIEW TOP		Panel No.	-
Panel Name	-				

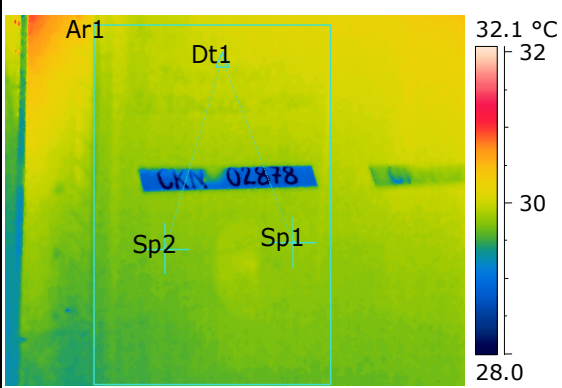
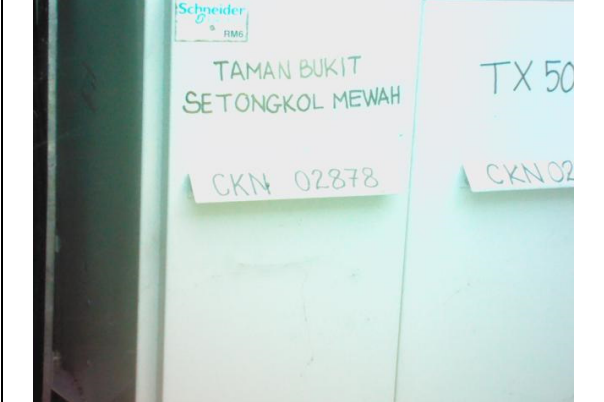
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	-	Serial No.	MA223850062
Spot Temp. 2 (Sp2)	-	Heater (A)	-
Average Temp. (Ar1)	30.7 °C	CB Position	-
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (ΔT)	-	Cable Type	-
Load (A)	-		
Defect	-		

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	MRMU	Manufacturer	SCHNEIDER	Time: 09:28 AM
	Model	RM6	Rating	12kV	Humidity: 77%
	Area	CABLE COMPARTMENT		Panel No.	CKN02878
Panel Name		TAMAN BUKIT SETONGKOL MEWAH			

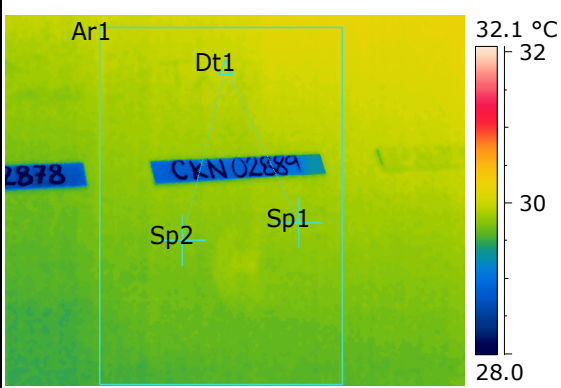
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PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	29.8 °C	Serial No.	MA223850062
Spot Temp. 2 (Sp2)	29.7 °C	Heater (A)	-
Average Temp. (Ar1)	29.8 °C	CB Position	CLOSE
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (Δ T)	0.1 °C	Cable Type	NOT ACCESSIBLE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	MRMU	Manufacturer	SCHNEIDER	Time: 09:28 AM
	Model	RM6	Rating	12kV	Humidity: 77%
	Area	CABLE COMPARTMENT		Panel No.	CKN02889
Panel Name		TX 500KVA			

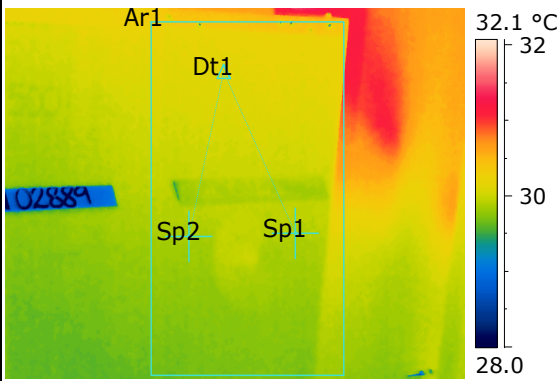
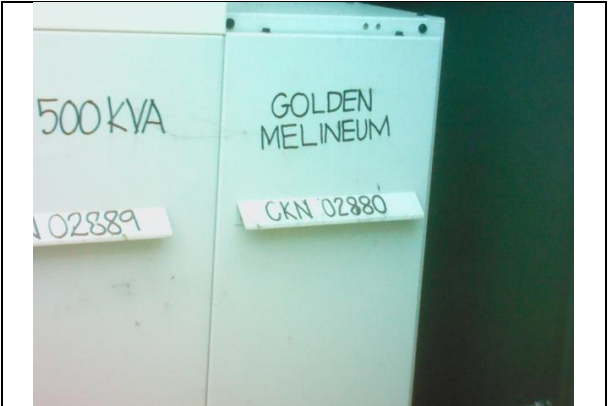
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	29.8 °C	Serial No.	MA223850062
Spot Temp. 2 (Sp2)	29.8 °C	Heater (A)	-
Average Temp. (Ar1)	29.8 °C	CB Position	CLOSE
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (ΔT)	0.1 °C	Cable Type	XLPE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	MRMU	Manufacturer	SCHNEIDER	Time: 09:28 AM
	Model	RM6	Rating	12kV	Humidity: 77%
	Area	CABLE COMPARTMENT		Panel No.	CKN02880
Panel Name		GOLDEN MELINEUM			

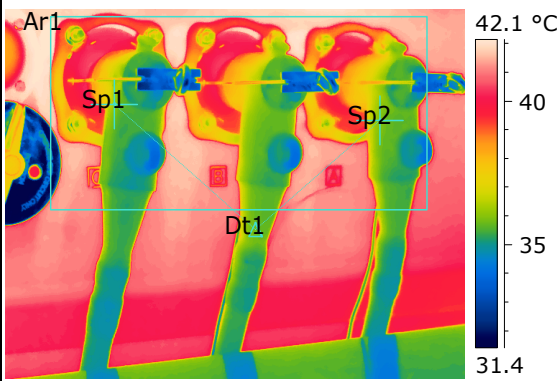
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	29.9 °C	Serial No.	MA223850062
Spot Temp. 2 (Sp2)	29.9 °C	Heater (A)	-
Average Temp. (Ar1)	30.0 °C	CB Position	CLOSE
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (Δ T)	0.0 °C	Cable Type	NOT ACCESSIBLE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	EWT	Time: 09:28 AM
	Model	HERMETICALLY SEAL	Rating	500KVA	Humidity: 77%
	Area	BUSHING		Tx No.	TX1
Location	HV SIDE				

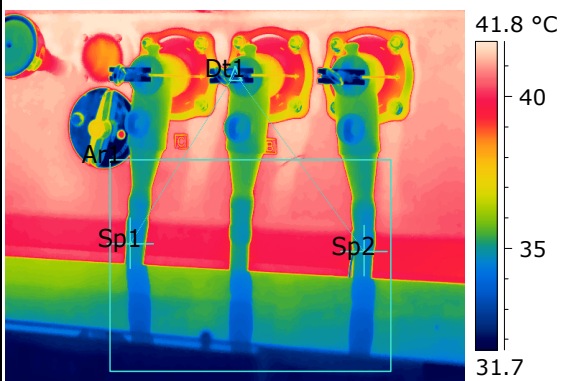
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	36.2 °C	Serial No.	EWT89428
Spot Temp. 2 (Sp2)	36.0 °C	Heater (A)	-
Average Temp. (Ar1)	38.3 °C	CB Position	-
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (ΔT)	0.2 °C	Cable Type	XLPE
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	TRANSFORMER	Manufacturer	EWT	Time: 09:28 AM
	Model	HERMETICALLY SEAL	Rating	500KVA	Humidity: 77%
	Area	CABLE		Tx No.	TX1
Location	HV SIDE				

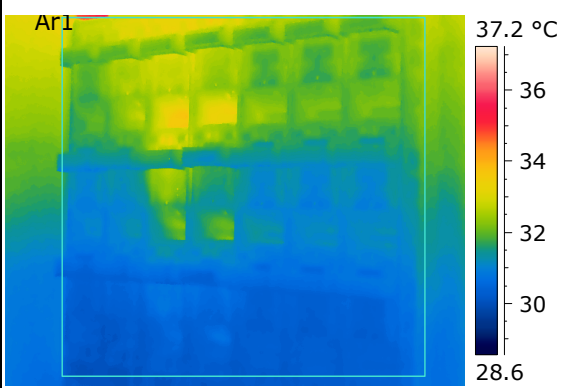
IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	34.9 °C	Serial No.	EWT89428
Spot Temp. 2 (Sp2)	34.6 °C	Heater (A)	-
Average Temp. (Ar1)	36.8 °C	CB Position	-
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (Δ T)	0.3 °C	Cable Type	XLPE
Load (A)	-		
Defect			

Analysis & Recommendations:

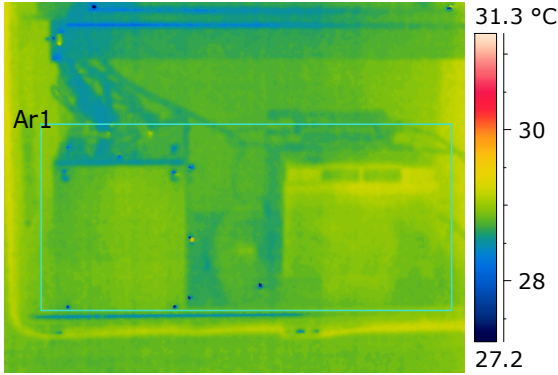
Analysis: No Anomaly.

Recommendation: -

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	LVDB TX1	Manufacturer	EWT	Time: 09:28 AM
	Model	DIN TYPE	Rating	800A	Humidity: 77%
	Area	OVERVIEW			Feeder No. -
Feeder Name		-			

IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	-	Serial No.	LVBCSU5287-500-11
Spot Temp. 2 (Sp2)	-	Heater (A)	-
Average Temp. (Ar1)	31.4 °C	CB Position	-
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (ΔT)	-	Cable Type	-
Load (A)	-		
Defect			
Analysis & Recommendations:			
Analysis: No Anomaly.			
Recommendation: -			

	Substation	CS LOT 7011 DARAT MAKBAR			Date: 03/03/2026
	Equipment	BATTERY BANK	Manufacturer	JUMBOHAN	Time: 09:28 AM
	Model	JLV-T-24-10/6	Rating	240V/50Hz	Humidity: 77%
	Area	OVERVIEW		Battery No.	1
Location	-				

IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)		Serial No.	121234-58
Spot Temp. 2 (Sp2)	-	Heater (A)	-
Average Temp. (Ar1)	28.9 °C	CB Position	-
Ambient Temp.	30.6 °C	Busbar Position	-
Delta T (ΔT)	-	Cable Type	-
Load (A)	-		
Defect			

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

SUBSTATION CONDITION

SUBSTATION OVERVIEW



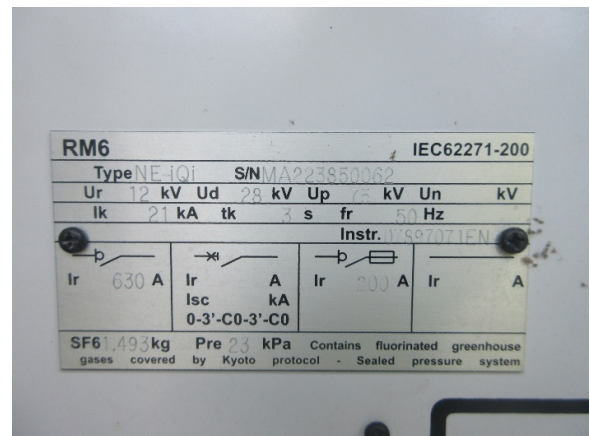
SIGNBOARD



SWITCHGEAR



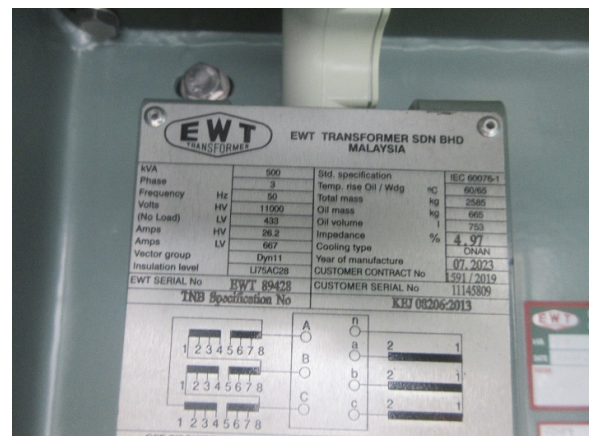
SWITCHGEAR NAMEPLATE



TRANSFORMER



TRANSFORMER NAMEPLATE



SUBSTATION CONDITION

LVDB



LVDB NAMEPLATE



EFI



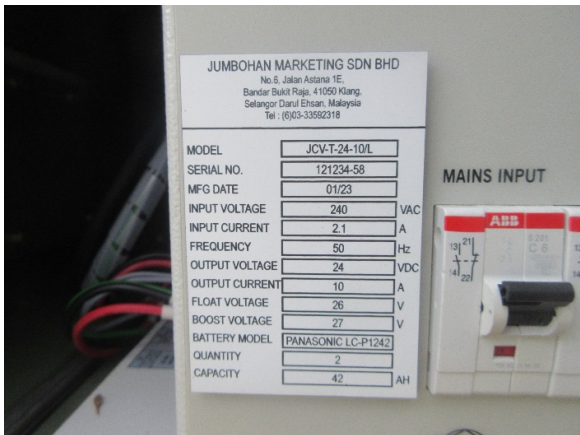
SF6 INDICATOR



BATTERY CHARGER



BATTERY CHARGER NAMEPLATE



SUBSTATION CONDITION

TRANSFORMER OIL LEVEL INDICATOR



VISUAL DEFECT

SUBSTATION



PADLOCK HOLDER BROKEN - LVDB TX1 BAY

TRENCHING



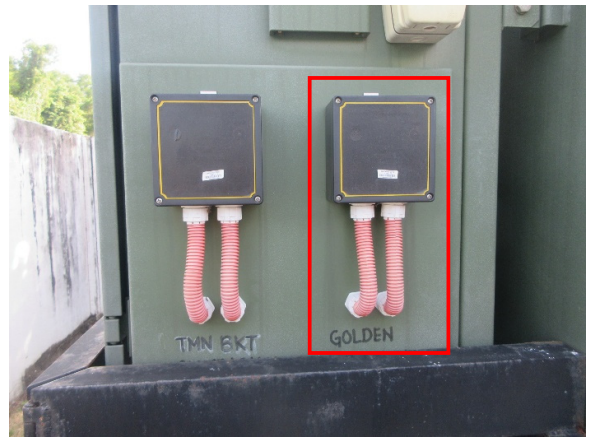
NO COVER – SWG BAY

RUBBER MAT



NO RUBBER MAT – SWG BAY

EFI



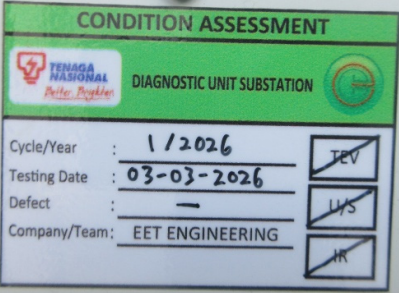
NO BATTERY SUPPLY – CKN02880 - GOLDEN
MELINEUM

FP/LVDB



NO FEEDER NO./FEEDER NAME – LVDB TX1 - NOT
FOLLOW NEW STANDARD

NORMAL/DEFECT STICKER

NORMAL STICKER	
	
CONDITION ASSESSMENT	
DIAGNOSTIC UNIT SUBSTATION	
Cycle/Year :	1 / 2026
Testing Date :	03-03-2026
Defect :	-
Company/Team :	EET ENGINEERING
	TEV
	U/S
	IR

156
CS

STATE / AREA: PAHANG / KUANTAN										DISTRIBUTION SUBSTATION (PCE) CMA TEST SHEET										DATE: 3/3/2026										FUNCTIONAL LOCATION: CKTN/PCE/201028																			
PE NAME: CS LOT 3011 DATAR MAYBAR										TIME IN / OUT: 9:28 A.M. / 9:57 A.M.										TEV BACK GROUND / HUMIDITY: 10 / 74										BACKGROUND TEMP: 30.6°C																			
V.I REF. NO:										SWITCHGEAR										TEMPERATURE										ULTRASOUND										TEV									
NO.	FDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE/SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLU)	COMP	Tmin (°C)	Tmax (°C)	AT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE / CYCLE	TYPE CHAR.	REMARK																												
1	CKN 02878	TAMAN BUKIT SETONKOL MEWAH	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE BREAKER TOP PANEL PT	29.8	29.8	0.0	29.8	106		-6	-	-	-	1	0	-																												
2	CKN 02889	Tx 500 KVA	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE BREAKER TOP PANEL PT	29.4	29.5	0.1	29.4	107		-6	-	-	-	9	0	-																												
3	CKN 02880	GOLDEN MELINEUM	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE BREAKER TOP PANEL PT	29.0	29.4	0.4	29.2	108		-6	-	-	-	10	0	-																												
4			TRIP / CLOSE		XLPE PILC N/A NACC			CABLE BREAKER TOP PANEL PT																																									
OVERVIEW																																																	
DETAILS										THERMOGRAPHY										ULTRASOUND										LOCAL TRANSFORMER																			
TX ID	SERIAL NO	CABLE TYPE/SIZE	REMARK	COMP.	Tmin (°C)	Tmax (°C)	AT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/SIZE	COMP.	Tmin (°C)	Tmax (°C)	AT	FILE	Avg (°C)	FILE	dB	TYPE CHAR.	REMARKS																									
TX1	EWT 89428	XLPE/PILC NACC XLPE/PILC		HT CABLE	34.2	34.7	0.5	34.6	111	-6	-	TX3		XLPE/PILC	HT CABLE																																		
				HT BUSHING	34.0	34.4	0.4	34.2	110	-6	-			XLPE/PILC	HT BUSHING																																		
				LV CABLE										XLPE/PILC	LV CABLE																																		
				LV BUSHING										XLPE/PILC	LV BUSHING																																		
				BODY										XLPE/PILC	BODY																																		
TX2		XLPE/PILC		HT CABLE								TX4		XLPE/PILC	HT CABLE																																		
				HT BUSHING										XLPE/PILC	HT BUSHING																																		
		XLPE/PILC		LV CABLE										XLPE/PILC	LV CABLE																																		
				LV BUSHING										XLPE/PILC	LV BUSHING																																		
				BODY										XLPE/PILC	BODY																																		
LV DISTRIBUTION BOARD / FEEDER PILLAR / OTHERS																																																	
1	LVDB / FP 1	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10																																			
	CABLE TYPE / SIZE	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC																																		
2	LVDB / FP 2	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10																																			
	CABLE TYPE / SIZE	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC																																		
NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin (°C)	Tmax (°C)	AT	FILE	Avg (°C)	REMARK																																									
	INC 1																																																
	INC 2																																																
	F1																																																
	F2																																																
	F3																																																
	F4																																																
	F5																																																
	F6																																																
	F7																																																
	F8																																																
	Battery Bank																																																
	Black Box 1																																																
	Black Box 2																																																
	Lightning Arrestor																																																
	Dropout Fuse																																																
	Auto Recloser																																																
	LBS																																																

Tested by (Name & Signature):

Tel: 603-5636 6205 Fax: 603-5636 7205

TNB Supervisor/Engineer (Name & Signature):

MOHD AZZAD BIN OTMAN (1055823)
JURUTEKNIK TINGKAT BIASA
DIAGNOSTIC SUBSTATION-PAHANG
SBU MAINTENANCE EAST
DISTRIBUTION NETWORK DIVISION

(156)

CS

VISUAL INSPECTION CHECKLIST FOR 11kV SUBSTATION

LVDB TXI (N-label), P-lock (holder br, LVDB TXI), NO MAT Bay 1

EFI (No battery supply, Panel 3)

Trench (N-Cover, sub bag)



DATE:

P/E Name (ERMS):

P/E Name (SITE):

GPS Coordinate:

PCE Type:

PCE Area:

Switchgear Type 1:

Manufacturer/Modal:

Switchgear Manufactured Date:

Switchgear Type 2:

Manufacturer/Modal:

Switchgear Manufactured Date:

No. of Transformer:

Tx 1:

Tx 2:

Tx 3:

Tx 4:

KEJANGGALAN

1. Signboard:
2. Earth connection:
3. EFI Functioning:
4. Heater Functioning:
5. Labelling @ Switchgear:
6. Labelling @ LVDB:
7. Gas Pressure:
8. Tx Guard:
9. Tx Oil Level (conservator tank):
10. Tx Oil Leak:
11. Tx Bushing:
12. Top Demand Indicator (@LVDB):
13. Link/fuse Holder:
14. LV Termination:
15. Rubber Mat:
16. Trenching:
17. Battery Voltage:
18. Lighting:
19. Roof Condition:
20. Fire Fighting:
21. Fence/door condition:
22. Cleanliness/Bushes:
23. Vandalism:
24. Other Findings

3 / 3 / 2026		CS LOT 7011 PARAT MAKBAR		F/L NO (ERMS):	CKTN / PCE / 1701028
3.809330, 103.318326		CS LOT 7011 PARAT MAKBAR		F/L NO (SITE):	CKTN / PCE / 1701028
INDOOR	☒	OUTDOOR	☐	COMPACT	☒
NORMAL	☒	INDUSTRIAL	☐	COASTAL	☐
RMU OIL	☐	RMU SF6	☒	MRMU	☒
SCHNEIDER / RM6		ATTACH BUILDING		PAT	
Switchgear Rating: 12 kV		VCB		OCB	
RMU SF6 / OIL S/N: MA223850062		OCB			
Switchgear Rating:		RMU SF6 / OIL S/N:			
Type		KVA	500	Construction Yr:	2023
Type		KVA		Manufacturer:	EWT
Type		KVA		Manufacturer:	
Type		KVA		Manufacturer:	
Type		KVA		Manufacturer:	
Type		KVA		Manufacturer:	
Type		KVA		Manufacturer:	
Type		KVA		Manufacturer:	

YES	☒	NO	☐
Good	☒	Not Good	☐
Good	☐	Not Good	☒
Good	☐	Not Good	☐
Good	☒	Not Good	☐
Good	☒	Not Good	☒
Good	☒	Not Good	☐
YES	☐	NO	☐
Good	☐	Not Good	☐
YES	☐	NO	☒
Good	☒	Not Good	☐
Good	☒	Not Good	☐
Good	☒	Not Good	☐
Good	☒	Not Good	☐
YES	☐	NO	☒
Good	☐	Not Good	☒
Good	☒	Not Good	☐
Good	☒	Not Good	☐
Good	☒	Not Good	☐
Valid	☒	Expired	☒
Good	☒	Not Good	☐
Good	☒	Not Good	☐
YES	☐	NO	☒

Remarks	
Remarks	
Remarks	NO SUPPLY
Remarks	N/A
Remarks	
Remarks	NO label
Remarks	
Remarks	N/A
Remarks	N/A
Remarks	
Remarks	
Remarks	
Remarks	
Remarks	
Remarks	NO MAT
Remarks	NO COVER
Remarks	
Remarks	
Remarks	
Remarks	
Remarks	Expiry Date: COMPACT SUB No. of missing portable: -
Remarks	
Remarks	
Remarks	

EET ENGINEERING SERVICES SDN BHD

(134556-W)

No. 38-2, 2nd Floor, Jalan 19/1D

47500 Subang Jaya, Selangor

Tel: 603-5636 6205 Fax: 603-5636 7205

TESTED BY (NAME & SIGNATURE):

TNB SUPERVISOR/ ENGINEER (NAME & SIGNATURE):

MOHD AZZAD BIN OTHMAN (10658230)
 JURUTEKNIK TINGGIATAN BIASA
 DIAGNOSTIC SUBSTATION-PAHANG
 SBU MAINTENANCE EAST
 DISTRIBUTION NETWORK DIVISION